

## Primer Aviso Design Decisions

### Design Decisions and Trade-offs

Build: Primer Aviso. Multi-agent first notice of loss for catastrophe surges. Microsoft Agent Framework (Python), Azure AI Foundry, gpt-5-mini (Data Zone Standard EU, Sweden Central).

Each entry records a decision made while building the reference implementation, the reasoning behind it, and the alternative that was considered and rejected.

1. Workflow architecture: six-step sequential workflow  
Decision: Build a six-step sequential workflow with a human gate at the end, rather than an open-ended agent.  
Rationale: First notice of loss has a fixed, known shape (safety check, intake, coverage, evidence, triage, approval). A fixed sequence is easier to test, explain, and audit than an agent choosing its own steps.  
Rejected: An autonomous agent deciding its own sequence of actions.
2. Safety gate: keyword-first, model-second  
Decision: Implement the safety gate as keyword-first, model-second.  
Rationale: Keyword match is instant, free, and cannot be talked around by phrasing. The model check only runs for wording the keyword list misses.  
Rejected: Relying on the model alone to detect a life-threatening emergency.
3. Coverage grounding: deterministic tool  
Decision: Use a deterministic tool returning exact clause text rather than a vector store for coverage grounding.  
Rationale: The household portfolio is 5 to 8 products. A deterministic tool returns exact clause text at zero cost, with no expiry risk.  
Rejected: Foundry file search. It adds cost per call, and vector stores expire after 7 days of inactivity, a real risk for a two-day build.
4. Clause checking: explicit earthquake-specific clause  
Decision: Require the coverage agent to check the earthquake-specific clause first, by name, before any general damage clause.  
Rationale: Live testing showed the model substituting a general damage clause for the earthquake-specific one, the single clause that actually matters for this event. The prompt now forces an explicit check, with the clause code cited.  
Rejected: Letting the model infer which clause applies from damage description alone.
5. Priority assignment: fixed rules in code  
Decision: Assign priority by code, from fixed rules, never by the model.  
Rationale: Priority drives handler triage order during a surge. It must be explainable and reproducible on demand.  
Rejected: Letting the intake or coverage agent classify priority based on its own judgment.
6. Photo filenames: assigned positionally by code  
Decision: Assign photo filenames by code, matched positionally, not trusted from the model's output.  
Rationale: The model receives only raw image bytes, with no filename attached, so

any filename it reports is invented. Live testing confirmed this: the model returned "foto\_1.jpg" instead of the real filename.

Rejected: Trusting the model to echo the correct filename in its structured output.

7. Coverage reporting: explicit not contracted statement  
Decision: Require the coverage agent to always state "not contracted" explicitly for a clause the household lacks.  
Rationale: A silent omission reads as a denial to a handler skimming quickly, and could be misread as the system having missed the check.  
Rejected: Listing only what is contracted and staying silent on what is not.
8. Related-report detection: flag, never merge  
Decision: Flag related reports from the same address, never merge them.  
Rationale: An apartment owner and a building administrator are legally distinct claims even at the same address. Merging them would risk losing one claim's record.  
Rejected: Automatically merging reports that share a normalised address.
9. Bereavement handling: separate branch and human routing  
Decision: Give bereavement a separate branch, tone, and field set, and route it to a person.  
Rationale: A death in the household changes what should be asked, and when. No automated system should make a condolence-adjacent decision.  
Rejected: Treating a death like any other data field in the standard intake flow.
10. Orchestration: local for the demo  
Decision: Run orchestration locally for the demo, not hosted.  
Rationale: This fits the two-day build timeline. Hosting is a known, separate step, explicitly logged as debt rather than skipped silently.  
Rejected: Deploying to Azure Container Apps or Functions before the interview.
11. Evidence photos: synthetic illustrations  
Decision: Use synthetic illustrations for evidence photos, not real photographs.  
Rationale: Sourced public-domain candidates looked like real press photojournalism, some showed identifiable real people, and a few risked resembling the actual August 2026 event. All three break the project's own synthetic-data rule.  
Rejected: Using real earthquake photos, even ones labelled public domain.